Lab 5 Access point

```
#include <ESP8266WiFi.h>
#include <ESP8266WebServer.h>

// Access Point credentials
const char* apSSID = "NodeMCU_AP";
const char* apPassword = "12345678";   // Minimum 8 characters

ESP8266WebServer server(80);

bool ledState = false;

void setup() {
  Serial.begin(9600);

  pinMode(LED_BUILTIN, OUTPUT);
  digitalWrite(LED_BUILTIN, HIGH);   // LED OFF initially

  // Start NodeMCU in Access Point mode
  WiFi.softAP(apSSID, apPassword);

  Serial.println();
  Serial.println("Access Point Started");
  Serial.print("SSID: ");
  Serial.println(apSSID);
  Serial.print("Password: ");
  Serial.println(apPassword);
  Serial.print("IP Address: ");
  Serial.println(WiFi.softAPIP());

  // Home page
  server.on("/", []() {
    server.send(200, "text/plain", "NodeMCU Access Point Web Server is Running!");
  });

  // Turn LED ON
  server.on("/on", []() {
    digitalWrite(LED_BUILTIN, LOW);   // Active LOW
    ledState = true;
    server.send(200, "text/plain", "LED is ON");
  });

  // Turn LED OFF
  server.on("/off", []() {
    digitalWrite(LED_BUILTIN, HIGH);
    ledState = false;
    server.send(200, "text/plain", "LED is OFF");
  });

  // Toggle LED
  server.on("/toggle", []() {
    ledState = !ledState;

    if (ledState) {
      digitalWrite(LED_BUILTIN, LOW);
      server.send(200, "text/plain", "LED is ON (Toggled)");
    } else {
      digitalWrite(LED_BUILTIN, HIGH);
      server.send(200, "text/plain", "LED is OFF (Toggled)");
    }
```

```
  });

  server.begin();
  Serial.println("HTTP server started");
}

void loop() {
  server.handleClient();
}
```